

BSDS (2025 Semester II): Statistics II

Midsem: Part A

NAME :	SOLUTION SET	ROLL :
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Time: 60 minutes

Total marks: 44

All the rough works should be done in the space provided in the question paper.

I. Answer all questions. Each question has one or more correct options. Identify all of them.

$6 \times 4 = 24$

1. Let $X_1, \dots, X_n$ be i.i.d. samples from $F_\theta \equiv \text{Uniform}(0, \theta)$. Indicate which of the following statements are correct?

A. $(\theta - \bar{X}_n)$ is an unbiased estimator of $\theta/2$.

☒

B. $\bar{X}_n$ is an unbiased estimator of $\theta/2$.

☒

C. Sample variance is the MoM estimator of the population variance.

☒

D. $\bar{X}_n^2/3$ is the MoM estimator of the population variance.

☒

Q1. A. $(\theta - \bar{X}_n)$ is NOT a statistic, hence NOT an estimator.

B. $E(\bar{X}_n) = E(X_1) = \int_0^\theta \theta^{-1} x dx = \theta/2$.

C. and D. By the MoM, we need to first estimate θ and then the MoM estimate of $g(\theta)$ would be $g(\hat{\theta}_{\text{MoM}})$. (*)

2. Let T_1, T_2 and T_3 be three estimators of $g(\theta)$.

A. If T_1 is unbiased for $g(\theta)$, then $\log T_1$ is unbiased for $\log g(\theta)$.

☒

B. If T_3 is an MLE for $g(\theta)$, then $\log T_3$ is an MLE for $\log g(\theta)$.

☒

C. If both T_1 and T_2 are unbiased for $g(\theta)$ then $(2T_1 + T_2)/3$ is also unbiased for $g(\theta)$.

☒

D. Among all the three estimators of $g(\theta)$, the one with minimum variance is the best.

☒

(*) Here population variance is $\theta^2/12$ and $\hat{\theta}_{\text{MoM}} = 2\bar{X}_n$.

Hence, MoM estimator of $\theta^2/12$ is $\bar{X}_n^2/3$.

Q2. A. $E[\log T_1] \neq \log E(T_1)$ in general.

B. By invariance property of MLE,

C. $E[(2T_1 + T_2)/3] = [2E(T_1) + E(T_2)]/3 = 3g(\theta)/3 = g(\theta).$

D. As $MSE(T, g(\theta)) = \text{Bias}^2(T, g(\theta)) + \text{Var}_\theta(T)$, estimator with (*)

3. Suppose you need to infer the population variance σ^2 , towards which you consider a random sample of size n , say, $X_1, \dots, X_n$. Indicate which of the following statements are correct.

A. Let T_n be an estimator of σ^2 with expectation $(n+2)\sigma^2/(n+1)$ and variance $\sigma^4/(n-1)^2$, and S_n be an unbiased estimator of σ^2 with variance $2\sigma^4/n^2$. Then S_n is a better estimator of σ^2 than T_n . ☒

B. Consider the estimator T_n described in part A. As n increases, the means of the sampling distributions of T_n necessarily become closer to σ^2 . ☒

C. Consider the estimator T_n described in part A. As n increases, the realizations of T_n necessarily become closer to σ^2 . ☒

D. Suppose you take another sample of same size, say, $Y_1, \dots, Y_n$, then the realizations of $\bar{Y}_n$ will necessarily be the same as the realization of $\bar{X}_n$. ☒

(*) minimum variance is NOT necessarily the one with lowest MSE.

Q3. A. $MSE(T_n, \sigma^2) = \left\{ \frac{n+2}{n+1} - 1 \right\}^2 \sigma^4 + \frac{\sigma^4}{(n-1)^2} = \sigma^4 \left\{ \frac{1}{(n+1)^2} + \frac{1}{(n-1)^2} \right\}$
 $= \frac{2(n^2+1)}{n^4-1} \sigma^4 = \frac{2}{n^2-1} \sigma^4$ and $MSE(S_n, \sigma^2) = \frac{2\sigma^4}{n^2}.$

So, clearly, S_n is better.

4. A person reaches his nearest bus-stop at 10 AM and waits for a particular bus. We are interested in studying the mean waiting time until the first bus arrives. Let X denote the waiting time (in minutes) for the person to catch the first bus. Determine which of the following distributions are appropriate for the model.

A. X follows a Geometric(θ) distribution with probability mass function

$$f_\theta(x) = \theta(1-\theta)^{x-1}, \quad x = 1, 2, \dots, \quad 0 < \theta < 1.$$

☒

B. X follows a Geometric(θ) distribution. with probability mass function

$$f_\theta(x) = \theta(1-\theta)^x, \quad x = 0, 1, \dots, \quad 0 < \theta < 1.$$

☒

C. X follows an Binomial(N, θ) distribution for some $N \in \mathbb{N}$.

☒

D. X follows a Poisson(θ) distribution.

☒

B. $E(T_n) - \sigma^2 = \frac{\sigma^2}{n+1}$. As $n \uparrow$, $|E(T_n) - \sigma^2| \downarrow$.

C. Not necessarily true. Let $T_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2$ and $\sigma^2 = 1$.

For realization $\{-1, 1\}$, $T_n - \sigma^2 = 1$. For the additional realization $\{-1, 1, 3\}$, $T_n - \sigma^2 = \frac{1}{3} \{4 + 4\} - 1 = \frac{5}{3} > 1$.

D. Realizations of sample mean differs as they are random. $\bar{X}_n$ and $\bar{Y}_n$ have same distribution, but the realizations

Can be different.

- Q4. A. The range of X excludes '0' and therefore is NOT an ideal model.
 C. There is no definite ending point N for waiting time. (X)

5. Let X be a random variable with the following distribution:

x	-3	-2	-1	0	1	2	3
$P(X=x)$	0.05	0.10	0.15	0.4	0.15	0.10	0.05

Consider the random variable $Y = \log |X|$. Indicate which of the following statements are true.

- A. $E(X^{21}) = 0$. ☒
 B. $E(Y^{21}) = -\infty$. ☒
 C. $P(Y = 0) = 0.15$. ☒
 D. $P(Y = 0) = 0.40$. ☒

(+) Hence $\text{Bin}(N, \theta)$ is not ideal.

- Q5. A. X is symmetric about 0. Hence $E(X^k) = 0$ for $k=1,3,\dots$
 B. Y takes the value $-\infty$ with prob. 0.4 and all other values of Y having +ve prob. mass are finite.
 So, $E(Y^k) = -\infty$ for any $k=1,3,\dots$

6. Let $X_1, \dots, X_{2n+1}$, $n \in \mathbb{N}$, be i.i.d. samples from $N(0,1)$ distribution. Indicate which of the following statements are correct:

- A. $\frac{X_1^2 + X_3^2 + \dots + X_{2n+1}^2}{X_2^2 + X_4^2 + \dots + X_{2n}^2} \sim F_{n,n}$. ☒
 B. $2^{-1}(X_1^2 + X_2^2 + \dots + X_{2n}^2) \sim \chi_n^2$. ☒
 C. $X_{2n+1} / \sqrt{X_1^2 + \dots + X_{2n}^2} \sim t_{2n}$. ☒
 D. $\frac{2nX_1^2}{X_1^2 + X_2^2 + \dots + X_{2n}^2} \sim F_{1,2n}$. ☒

C. and D. $P(Y \neq 0) = P(|X|=1) = P(X=1) + P(X=-1) = 0.3$

- Q6. A. $T_1 = X_1^2 + X_3^2 + \dots + X_{2n+1}^2 \sim \chi_{n+1}^2$ and $T_2 = X_2^2 + \dots + X_{2n}^2 \sim \chi_n^2$,

$T_1 \perp T_2$. So, $F = \frac{T_1/n+1}{T_2/n} \sim F_{n+1,n}$. $\therefore \frac{T_1}{T_2} \sim \frac{n+1}{n} F_{n+1,n} \neq F_{n,n}$

- B. $T_3 = X_1^2 + \dots + X_{2n}^2 \sim \chi_{2n}^2 \equiv \text{Gamma}(n, \frac{1}{2})$ and [P.T.0]
 $T_3/2 \sim \text{Gamma}(n, 1) \neq \chi_n^2$ (verify)

- C. As $X_{2n+1} \perp T_3$, $T = X_{2n+1} / \sqrt{T_3/2n} \sim t_{2n}$ but
 $T/\sqrt{2n} \not\sim t_{2n}$.

D. $X_1^2 \not\sim T_3$. So, $X_1^2 / (T_3/2n) \not\sim F_{1,2n}$.

II. Write the answers to the following questions in one sentence.

$5 \times 4 = 20$

1. Let $X_1, \dots, X_n$ be i.i.d. random variables from the inverse Gaussian distribution with parameters $\theta = (\mu, \lambda)$ and common pdf

$$f_{\theta}(x) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp \left\{ -\frac{\lambda(x - \mu)^2}{2\mu^2 x} \right\}, \quad x > 0, \mu > 0, \lambda > 0.$$

Then a complete and sufficient statistic for θ is:

The joint pdf of $X_1, \dots, X_n$ is:

$$f_{\theta}(x) = \exp \left\{ \frac{n}{2} \log \lambda - \frac{n}{2} \log (2\lambda) - \frac{3}{2} \sum_{i=1}^n \log x_i - \frac{\lambda}{2\mu^2} \sum_{i=1}^n x_i + \frac{\lambda n}{\mu} - \frac{\lambda}{2} \sum_{i=1}^n \frac{1}{x_i} \right\}$$

Thus, $\left\{ \sum_{i=1}^n x_i, \sum_{i=1}^n \frac{1}{x_i} \right\}$ is jointly complete and sufficient for θ .

2. Let $X_1, \dots, X_n$ be i.i.d. random variables from a distribution with the following p.d.f.

$$f(x) = \begin{cases} (2x^3)^{-1} & \text{for } x \geq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

Then the pdf of $Y = X_{(1)}$ is:

$$P(Y \geq y) = \{P(X_1 \geq y)\}^n = \left[\int_y^{\infty} \frac{1}{2x^3} dx \right]^n = \left\{ \left[-\frac{x^{-2}}{4} \right]_y^{\infty} \right\}^n = \left(\frac{1}{2y} \right)^{2n} \text{ for } y \geq 1/2 \text{ and } P(Y \geq y) = 1 \text{ for } y < 1/2.$$

$$\text{So, } F_Y(y) = 1 - \left(\frac{1}{2y} \right)^{2n} \text{ for } y \geq 1/2 \text{ and } F_Y(y) = 0 \text{ otherwise.}$$

[ANS] So, PDF of Y is $f_Y(y) = \frac{2n}{y^{2n+1}} \text{ for } y \geq 1/2 \text{ and } f_Y(y) = 0$

otherwise.

3. Let $X \sim \text{Uniform}(0, 10.5)$. Let $Y = \lfloor X \rfloor$, where $\lfloor a \rfloor$ is the greatest integer not exceeding a . Then $E(Y)$ is (write in mixed fraction):

$X \sim \text{Uniform}(0, 10.5)$ and $Y = \lfloor X \rfloor$. So, Y has the prob. mass in $\{0, 1, \dots, 10\}$.

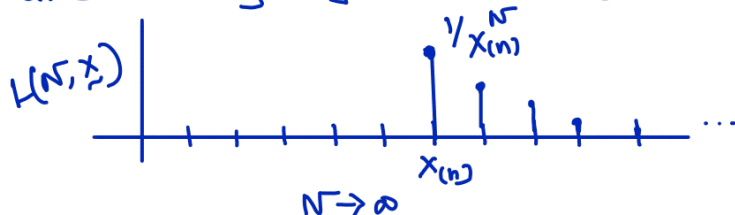
Now, $P(Y=k) = P(X \in [k, k+1)) = \int_k^{k+1} \frac{dx}{10.5} = \frac{1}{10.5} = \frac{2}{21}$
and $P(Y=10) = \int_{10}^{10.5} \frac{dx}{10.5} = \frac{0.5}{10.5} = \frac{1}{21}$ for $k=1, \dots, 9$.

[ANS] So, $E(Y) = \frac{2}{21} (1+2+\dots+9) + \frac{10}{21} = \frac{1}{21} \{9 \times 10 + 10\} = \frac{100}{21} = 4 \frac{16}{21}$

4. Let $X_1, \dots, X_n$ be i.i.d. random variables from the discrete uniform distribution supported on $\{1, \dots, N\}$, $N \in \mathbb{N}$. Then the MLE of N is :

The joint pmf of $X_1, \dots, X_n$ is $P(X_1=x_1, \dots, X_n=x_n) = \frac{1}{N^n}$

The likelihood fn. of N , $L(N, \underline{x})$ is: for $X_{(1)} \geq 1$ and $X_{(n)} \leq N$.



So, MLE of N is $X_{(n)}$. [ANS]

5. Let S_n be an unbiased estimator of $\sigma^2 \in (0, \infty)$ with variance $8\sigma^4/\{5(n-1)^2\}$. The minimal sample size n such that the ratio S_n/σ^2 lies between $[1/2, 3/2]$ with probability at least 0.9 is :

Require minimum choice of n such that

$$P\left(\frac{1}{2} \leq \frac{S_n}{\sigma^2} \leq \frac{3}{2}\right) \geq 0.9 \Rightarrow P\left(-\frac{1}{2} \leq \frac{S_n}{\sigma^2} - 1 \leq \frac{1}{2}\right) \geq 0.9$$

$$\Rightarrow P\left(\left|\frac{S_n}{\sigma^2} - 1\right| \leq \frac{1}{2}\right) \geq 0.9. \text{ By Chebyshev's inequality}$$

$$P\left(\left|S_n - \sigma^2\right| > \frac{\sigma^2}{2}\right) \leq \frac{\text{Var}(S_n)}{\sigma^4/4} = \frac{8\sigma^4 \times 4}{5(n-1)^2 \sigma^4} = \frac{32}{5(n-1)^2}$$

$$\therefore P\left(\left|\frac{S_n}{\sigma^2} - 1\right| \leq \frac{1}{2}\right) \geq 1 - \frac{32}{5(n-1)^2}. \text{ So, enough to choose}$$

$$n \text{ such that } 1 - \frac{32}{5(n-1)^2} \geq 0.9 \Rightarrow \frac{1}{10} \times 5(n-1)^2 \geq 32$$

$$\Rightarrow (n-1)^2 \geq 64 \Rightarrow n \geq 8+1=9.$$

[ANS] Minimum possible sample size ensuring $P\left(\frac{1}{2} \leq \frac{S_n}{\sigma^2} \leq \frac{3}{2}\right) \geq 0.9$ is 9.

Rough work